

Table 2: A comparison of selected AWS, Azure and GCP cloud-based AI services

Service category	AWS	Azure	GCP
Pre-trained AI services	AWS offers a range of pre-trained AI services for various domains, such as vision, speech, language, chatbots, forecasting, and recognition. Examples are Amazon Rekognition, Amazon Polly, Amazon Comprehend, Amazon Lex, Amazon Forecast, and Amazon Rekognition.	Azure offers a range of pre-trained AI services for various domains, such as vision, speech, language, decision, and web search. Some examples are Azure Cognitive Services, Azure Speech Services, Azure Language Understanding, Azure Personalizer, and Azure Cognitive Search.	GCP offers a range of pre-trained AI services for various domains, such as vision, speech, language, translation, natural language, video, and recommendations. Examples include Google Cloud Vision API, Google Cloud Speech-to-Text, Google Cloud Translation, Google Cloud Natural Language, Google Cloud Video Intelligence, and Google Cloud Recommendations AI.
Custom AI services	AWS's range of custom AI services allow users to build, train, and deploy their own machine learning models. Examples are Amazon SageMaker, Amazon Kendra, Amazon Textract, Amazon Fraud Detector, and Amazon Augmented AI.	Azure's custom AI services also allow users to build, train, and deploy their own machine learning models. Some examples are Azure Machine Learning, Azure Form Recognizer, Azure Anomaly Detector, Azure Synapse Analytics, and Azure Machine Learning Designer.	GCP, too, offers a range of custom AI services that allow users to build, train, and deploy their own machine learning models. Examples include Google Cloud AI Platform, Google Cloud AutoML, Google Cloud Document AI, Google Cloud Dataflow, and Google Cloud Vertex AI.
Data analytics and processing	AWS's data analytics and processing services enable users to collect, store, analyse, and visualise large-scale data. Examples are Amazon Kinesis, Amazon Redshift, Amazon Athena, Amazon EMR, and Amazon QuickSight.	Azure also offers a range of data analytics and processing services that enable users to collect, store, analyse, and visualise large-scale data. Some examples are Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure HDInsight, and Azure Power BI.	GCP's data analytics and processing services, too, enable users to collect, store, analyse, and visualise large-scale data. Examples are Google Cloud Pub/Sub, Google BigQuery, Google Cloud Dataflow, Google Cloud Dataproc, and Google Data Studio.

and comprehensive security features and policies of Azure, which can protect your data and resources from unauthorised access and threats.

Cost-effectiveness and efficiency:

The cost-effectiveness and efficiency of your AI solutions can be optimised using the pay-as-you-go and consumption-based pricing models of Azure, which can help you save money and resources.

Table 2 compares some of the AI services offered by AWS, Azure, and GCP.

AWS AI services

Amazon has emerged as a trailblazer with its comprehensive suite of AI services. Harnessing the power of advanced machine learning algorithms,

scalable cloud infrastructure, and deep integration capabilities, Amazon AI services deliver robust solutions for complex problems.

Amazon SageMaker

Amazon SageMaker is a fully managed machine learning service that enables developers and data scientists to build, train, and deploy machine learning models at scale. SageMaker provides a suite of tools including managed Jupyter Notebooks for exploratory data analysis, built-in high-performance algorithms, and support for custom model development with Docker containers. It offers distributed training and hyperparameter optimisation, and allows seamless deployment of models to production for real-time and batch

inference. SageMaker also includes features such as SageMaker Ground Truth for data labelling and SageMaker Studio, an integrated development environment for machine learning.

Amazon Rekognition

Amazon Rekognition is a computer vision service that analyses images and videos to identify objects, people, text, scenes, and activities. It leverages deep learning models to provide functionalities like facial analysis, celebrity recognition, and detection of explicit content. Rekognition can perform real-time analysis on video streams and supports custom labels for training custom models. It integrates with other AWS services like Lambda and S3, making it suitable for security surveillance, automated content